



SECTION 3: Executive Analytics Framework — The 4 Grand Pillars

Plaintext



PHA-WAW-001 Master Analytical Pillars

- Pillar 1: Clinical Outcomes & Care Continuity
- Pillar 2: Operational Efficiency & Capacity Optimization
- Pillar 3: Financial Security & Revenue Cycle Management
- Pillar 4: Social Determinants of Health (SDOH) & Vulnerability



PILLAR 1: Clinical Outcomes & Care Continuity

Strategic Focus: Evaluate clinical trajectories, psychopharmacological efficacy, adverse event mitigation, and longitudinal recovery stabilization.

Analytical Module 1.1: Psychopharmacological Adherence vs. Crisis Recurrence

- **Business Hypothesis:** Sub-optimal adherence rates ($\text{Adherence_Rate} < \tau$) correlate directly with escalated crisis events (ER_Visit_Flag , $\text{Hospitalization_Flag}$) and reduced overall time-to-recurrence.
- **Analytical Causality:** Instability in serum/psychopharmacological compliance leads to neurochemical destabilization, escalating acute psychiatric decompensation requiring emergency interventions.
- **Target Schema:** `Medication_History` , `Crisis_Events` , `Patients` .
- **Expected Stakeholder Impact:** Establishes precise clinical adherence thresholds for early pharmacist/psychiatrist outreach, reducing emergency department over-utilization and unplanned acute inpatient admissions.

Analytical Module 1.2: Medication Alterations, Adverse Side Effects, and Interim Progression

- **Business Hypothesis:** Severe adverse side effects ($\text{Side_Effect_Severity}$) combined with frequent regimen changes ($\text{Medication_Change_Flag}$) impair interim clinical score improvements ($\text{Interim_Clinical_Score}$) during psychotherapy sessions.

- **Analytical Causality:** High adverse event burden compromises patient cognitive and emotional capacity, reducing engagement in psychotherapeutic modalities and stalling clinical progression.
- **Target Schema:** Treatments , Longitudinal_Sessions , Outcomes .
- **Expected Stakeholder Impact:** Identifies high-risk pharmacological regimens to optimize dosage titration and balance efficacy against side-effect burden.

Analytical Module 1.3: Unstructured Sentiment vs. Structured Risk Discordance

- **Business Hypothesis:** Negative sentiment shifts in clinician notes (Sentiment_Score , Self_Harm_Mentions , Hallucination_Mentions) manifest prior to formal updates in structured risk assessment scores (Suicide_Risk_Score).
- **Analytical Causality:** Qualitative behavioral indicators captured in narrative notes act as leading indicators of clinical deterioration before periodic structured scoring tools are formally re-administered.
- **Target Schema:** Clinical_Notes , Clinical_Assessments , Patients .
- **Expected Stakeholder Impact:** Bridges narrative clinical documentation with quantitative risk monitoring to catch impending crises early.

Analytical Module 1.4: Digital/Behavioral Patient Engagement vs. Trajectory Outcomes

- **Business Hypothesis:** High homework completion rates (Homework_Completed) and active mobile application usage (App_Usage_Hours) significantly increase the probability of a positive outcome trajectory (Trajectory_Outcome).
- **Analytical Causality:** Active self-management and out-of-session behavioral reinforcement accelerate cognitive restructuring and long-term symptom management.
- **Target Schema:** Longitudinal_Sessions , Outcomes .
- **Expected Stakeholder Impact:** Quantifies the ROI of digital mental health tools and home therapy integration, supporting evidence-based digital health investments.



PILLAR 2: Operational Efficiency & Capacity Optimization

Strategic Focus: Eliminate operational bottlenecks, reduce appointment non-attendance, optimize facility throughput, and manage workforce burnout and continuity.